

NEWS

BIOMEDICINE

Experimental treatments target the brain's 'plumbing'

Animal studies—and controversial human surgeries—aim to boost fluid clearance to fight neurological disease **JENNIE ERIN SMITH**



Bolstering glymphatic pathways, shown in the brain of a mouse, is a possible strategy for treating Alzheimer's and other diseases.

More than a decade ago, when researchers discovered a ghostly network of microscopic channels that push fluid through the brain, they began to wonder whether the brain's plumbing, as they sometimes refer to it, might be implicated in neurodegenerative diseases such as Alzheimer's. Now, they are testing a host of ways to improve it.

At the Society for Neuroscience (SfN) meeting last month in San Diego, several teams reported early promise for drugs and other measures that improve fluid flow, showing they can remove toxic proteins from animal or human brains and reverse symptoms in mouse models of neurological disease.

Plastic surgeons in China, meanwhile, have gone further, conducting experimental surgeries that they say help flush out disease-related proteins in people with Alzheimer's. The trials have generated excitement but also concern over their bold claims of success. A group of academic surgeons in the United States is planning what they say will be a more rigorous clinical trial, also in Alzheimer's patients, that could be-

gin recruiting as early as next year.

The surgical approach "sounds unbelievable," says neuroscientist Jeffrey Iliff of the University of Washington. "But I'm not going to say I know it can't work. Remember, 13 years ago we didn't know any of this existed."

In 2012, Iliff, with pioneering Danish neuroscientist Maiken Nedergaard and colleagues, described a previously unrecognized set of fluid channels in the brain that they dubbed the glymphatic system. Three years later, other groups revealed a second, related system of fluid transport: a matrix of tiny lymphatic vessels in the meninges, or membranes covering the brain.

This early work showed the glymphatic channels form in the spaces around blood vessels, whose rhythmic contractions and dilations help push cerebrospinal fluid (CSF) through the channels. Traveling into the brain along arteries, the fluid picks up waste from deep inside brain tissue, then flows along veins to the meningeal lymphatic vessels. From there it travels to the lymph nodes of the neck, finally exiting to the bloodstream.

Brain clearance researchers have

learned this system is most active during deep sleep, and that poor sleep, aging, traumatic brain injury (TBI), and cerebrovascular disease all harm it. They have begun to link impaired brain clearance to a higher risk of developing dementia, and scientists think it may be the mechanism by which poor sleep raises dementia risk. Iliff's group collaborated on a study, reported in an October preprint, that found the system flushes two proteins implicated in Alzheimer's, amyloid and tau, out of the human brain.

"We think [fluid clearance] has implications not just for Alzheimer's, Huntington's, and Parkinson's, but also stroke and headache and traumatic brain injury and probably mood disorders," Iliff says.

The research has already inspired wellness enthusiasts to tout various unproven brain-cleansing methods, ranging from neck yoga to sleeping on one's side. But researchers, too, are looking seriously at ways to improve brain clearance. One is to fortify the meningeal lymph vessels, an approach tested by neurobiologist Wenzhen Duan of Johns Hopkins University and her colleagues. They worked with mice that have the genetic mutation that causes Huntington disease, and show movement symptoms similar to those seen in people.

At the SfN meeting, Duan's group reported that injecting a protein called vascular endothelial growth factor C, which helps grow and maintain lymph vessels, flushed the toxic huntingtin protein out of the mice's brains and improved their motor coordination. The team also explored a second brain-clearing strategy targeting Aquaporin-4, a protein expressed in brain cells called astrocytes, which give glymphatic channels their structure. Using a gene therapy to increase expression of Aquaporin-4 improved clearance and reduced huntingtin levels in the brain, they found.

Tweaking vasomotion, the subtle pulsations that drive CSF through glymphatic channels, might also be beneficial, according to neuroscientist Rashad Hussain of the University of Maryland. The elevated levels of the neurotransmitter norepinephrine seen after brain injury in mice and people are known to stymie glymphatic flow. Administering a cocktail of three drugs known

as alpha and beta blockers improved the mice's brain fluid clearance immediately after their injuries by boosting vasomotion, Hussain and his colleagues found. At the SfN meeting, Hussain reported the same drugs also normalized the animals' sleep patterns, which were disrupted after injury.

The cocktail might be used to treat both acute TBI and the lingering sleep problems it can cause, Hussain says—and ultimately might help reduce downstream consequences. “TBI is also a risk factor for Alzheimer's disease, which we know does not happen overnight—it takes years to develop, even decades,” he says. “So if we can improve clearance when there is a sleep disturbance, maybe we can slow down the progression.”

Mechanical interventions have also been shown to jog brain clearance. In June, a group led by vascular scientist Gou Young Koh of the Korea Advanced Institute of Science and Technology reported that stimulating the lymph nodes around the jaw with a pulsating device restored lymphatic flow from the brains of aged mice.

At the SfN meeting, Mitchell Bartlett, a brain lymphatics researcher at the University of Arizona, presented evidence from mice that such interventions might help treat Alzheimer's. **Two months of twice-daily massage of the head and neck in mice engineered to develop Alzheimer's-like symptoms improved the animals' cognitive function, reduced amyloid in the brain, and lowered blood biomarkers of neurodegeneration, reported Bartlett, who also published the results in an August preprint. He hopes to launch a clinical trial of the intervention.**

Neuropsychologist Sephira Ryman of the University of New Mexico presented findings published last month showing older adults and people with Parkinson's disease could increase brain vasomotion and cerebrospinal fluid flow by breathing dilute carbon dioxide, which dilates blood vessels, for short periods. The intervention also seemed to sweep proteins and peptides linked to neurodegeneration and inflammation out of the brain and into the blood. “We're really encouraged by these two findings,” Ryman says, adding that the next step is to measure clinical effects.

None of these strategies is nearly as daring—or invasive—as the microsurgical procedures developed in China as a treatment for Alzheimer's. The surgery, called deep cervical lymphovenous anastomosis (dcLVA), is an adaptation of a common treatment for lymphedema, or swelling in the body caused by blocked lymphatic flow. It connects lymph nodes or lymphatic vessels in the neck to veins in the hope of boosting brain fluid clearance.

In 2020, plastic surgeon Qingping Xie of Hangzhou Qiushi Hospital was the first to perform the procedure in an Alzheimer's patient. Xie now advertises the surgeries online to people in China and overseas, urging them to “Stop progressing, Start recovering!” A meta-analysis, released in July as a preprint, reported that more than two dozen trials of dcLVA procedures, enrolling some 1300 Alzheimer's patients, are ongoing at centers across China.

The largest dcLVA study published to date came out in October: Alzheimer's researcher Jianping Ye and his colleagues at Zhengzhou University described results from 41 Chinese patients with mild to moderate Alzheimer's who underwent surgical treatment. As a surrogate measure of brain fluid clearance, the researchers measured blood and CSF levels of amyloid and tau, which showed improvement in two-thirds of patients 3 months after surgery. Ye says these changes have persisted in later follow-up tests. He and his colleagues also reported cognitive improvements. “This is the most effective therapy for [Alzheimer's], to my knowledge,” Ye says.

The surgeries have faced criticisms, including from neurosurgeons in China citing a lack of solid evidence, and some experts are skeptical of the idea that glymphatic impairment is a type of brain lymphedema you can unclog. It “doesn't work literally like plumbing,” Duan says, because much of the malfunction is occurring inside the astrocytes. Roslyn Bill, a brain clearance researcher at Aston University, says she, too, doubts much glymphatic function can be regained by opening the tap in the neck. “Mechanical processes may well cause all sorts of cellular pathways to be affected,” she says, but no one has yet shown how.

Others are intrigued. Plastic surgeon Bohdan Pomahac of Yale

IN OTHER NEWS

SHARKS, RAYS GAIN PROTECTION

Global sales of many endangered sharks and rays are expected to be banned by the Convention on International Trade in Endangered Species of Wild Fauna and Flora (CITES), which concludes its triennial conference of parties this week in Uzbekistan. The move would protect various species that have been severely depleted by the trade in their fins, meat, and oil. International trade will be banned for the critically endangered oceanic whitetip shark and whale shark, 10 species of manta and devil rays, and 16 species of sharklike rays. Other species—smoothhound sharks, the tope shark, and gulper sharks—will be added to a list that allows for international trade but requires permits. The Wildlife Conservation Society called the development “the most comprehensive advance in global shark conservation in CITES history.” —Erik Stokstad

BATS HIT BY BIRD FLU In late 2022, an unprecedented outbreak of the avian influenza strain H5N1 reached South America, ravaging not only birds, but also marine mammals along the Pacific coast. Now, a preprint posted on bioRxiv last month reports the massive outbreak may have also affected vampire bat populations in coastal Peru that fed on the blood of infected marine animals. The virus—identified based on antibodies in the bats' blood—did not appear to spread between bats, but if it did, the animals could become a new reservoir, the researchers warn. And because some vampire bats also feed on livestock, they could become a bridge spreading the virus between marine and terrestrial animal populations. Another team had previously found H5N1 in flying foxes, a bat species that feeds on fruit and nectar, in Bangladesh. Those bats were likely infected by sharing a tree with house crows battered by the virus. —Martin Enserink

University remembers attending a presentation by Xie and his colleagues at surgical meeting a few years ago and finding it “bizarre.” The protocols of the studies in China were “all over the map,” he says, and even the published studies “were pretty poor quality and low impact.” But as promising results continued to trickle in, Pomahac put together a multidisciplinary team to investigate further. Now, he is working closely with researchers at Washington University in St. Louis who are conducting dcLVA surgeries in animals.

If all goes well and federal funding comes through this month, Pomahac hopes to launch a clinical trial of an dcLVA surgery in carefully screened patients in the early stages of Alzheimer’s. “No matter how unlikely it is [to work], it’s worth exploring properly and studying scientifically,” he says.

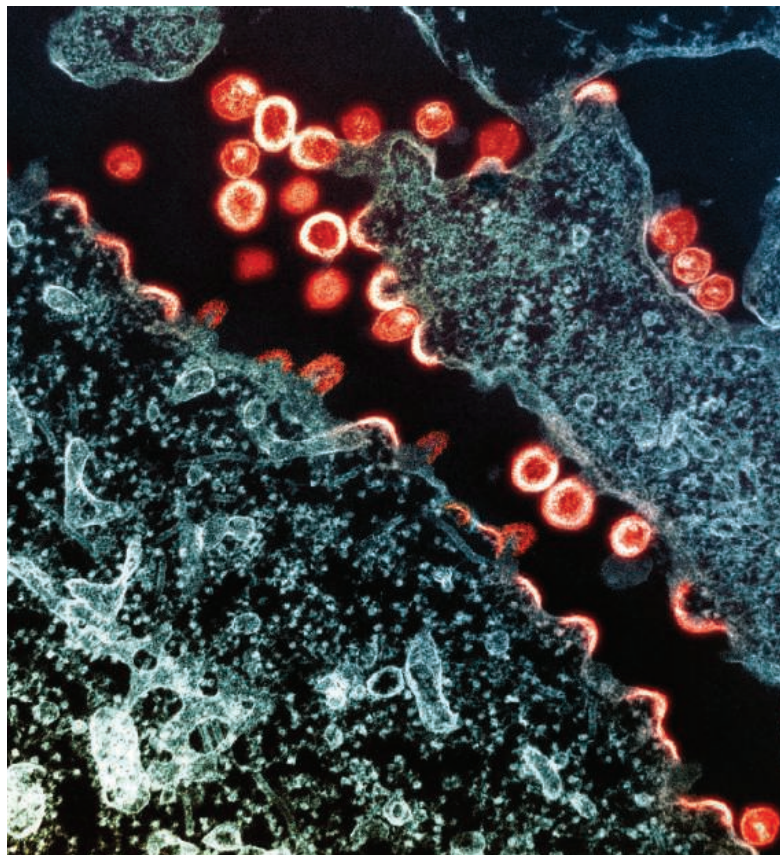
Bill says she would not undergo an dcLVA surgery—at least not without stronger evidence. She would, however, take a pill to keep her glymphatic system running smoothly as she ages.

In 2020, Bill reported that an antipsychotic called trifluoperazine, which prevents Aquaporin-4 from being shifted around within cells, prompted a dramatic recovery in a mouse model of CNS edema, or swelling in the brain and spinal cord, caused by injury. The mice behaved “as though they’d never been injured,” she says.

Trifluoperazine “is an old, dirty drug,” Bill acknowledges. “But if we could test it today, I think we would be in a position where we could save people’s lives.”

Iliff has also been studying an old drug—the alpha blocker prazosin, which his group found in a retrospective study to be potentially helpful in preventing dementia among veterans taking it for trauma-related nightmares. His team is testing its effects on glymphatic function and markers of Alzheimer’s in rodents and nonhuman primates. (Prazosin is one of the three drugs in Hussain and colleagues’ experimental cocktail.) Iliff is also working with the California biotech firm Applied Cognition on a noninvasive device to measure glymphatic clearance in real time.

“There is no direct evidence yet that lack of glymphatic efficiency leads to dementia in people,” Bill points out. “But the circumstantial evidence is strong enough that it’s worth a shot to drug it.” □



INFECTIOUS DISEASES

‘Functional cure’ for HIV comes closer to reality

Studies finger specific immune response that helps some people control the virus without drugs **JON COHEN**

Of the estimated 91 million people infected with HIV over the past 45 years, no one has fully defeated the virus, save for a dozen people who needed risky stem cell transplants to cure blood cancers. But a small percentage of patients in experimental trials have achieved a “functional cure,” in which the immune system contains the virus and patients can stop taking antiretroviral drugs (ARVs) for many months, or even years.

Now, two independent research groups have shown that one particular class of immune cells appears to play a crucial role in these functional cures. Finding ways to nurture those cells might make it possible to free more patients from the need to take drugs for life.

Those cells “may be the magical biomarker we need for a cure,” says Steve Deeks, an HIV clinician at the University of California, San Francisco (UCSF) who co-led one of the studies, which *Nature* published this week on the occasion of World AIDS Day. “It’s promising for a future in which you can intervene and tilt a whole series of events back in the direction of the immune system,” says Dennis Burton, an immunologist at Scripps Research who was not involved with the studies.

ARVs powerfully knock back levels of HIV in both blood and tissues, but stubborn “reservoirs” of cells that harbor dormant viral DNA persist in the body. When people stop taking drugs, the virus almost always resurges within weeks.

In a functional cure, the immune